

DELHI PUBLIC SCHOOL
ANNUAL EXAMINATION 2026 — SCIENCE

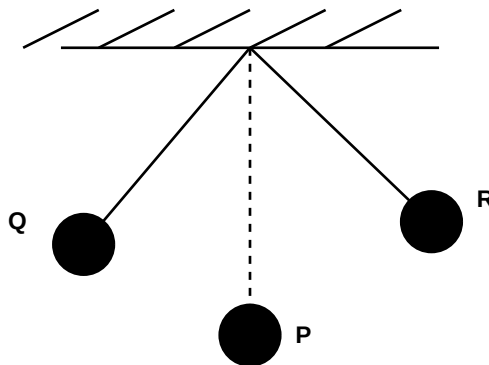
Class X | Time: 3 Hours | Maximum Marks: 80

General Instructions:

1. This paper contains two sections: Section A and Section B.
2. All questions are compulsory unless a choice is indicated.
3. Marks for each question are indicated against it, where printed.
4. Draw diagrams wherever necessary and label them clearly.

SECTION A — PHYSICS

1. Define the term 'inertia'. [2 marks]
2. State the SI unit of force and define one newton. [2 marks]
3. Solve for x: $2x^2 - 5x + 3 = 0$. Show all steps of your working. [5 marks]
4. The diagram below shows a simple pendulum released from rest at point R and allowed to swing through its lowest point P. Label points P, Q and R, and explain the energy transformation of the bob as it moves from R to P. [5 marks]



Answer either Question 5 or Question 6.

5. State and explain Newton's second law of motion, with one real-life example. [5 marks]
6. Derive the expression for the kinetic energy of a body of mass m moving with velocity v . [5 marks]
7. Why does ice float on water? Explain in terms of density.
8. Specific heat and heating of water.
(a) Define specific heat capacity. [2 marks]

(Section A continues on the next page)

- (b)** Calculate the heat required to raise the temperature of 2 kg of water from 20 degrees Celsius to 80 degrees Celsius. (Specific heat capacity of water = 4200 J/kg-degC) [3 marks]

SECTION B — BIOLOGY & CHEMISTRY

- 1.** Name the powerhouse of the cell. [1 marks]
- 2.** What is photosynthesis? Write the balanced chemical equation for photosynthesis. [4 marks]
- 9.** List two differences between plant cells and animal cells. [4 marks]
- 10.** Balance the following chemical equation: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$ [3 marks]
- 11.** Acids, bases and ionic conduction.
 - (a i)** What is the pH of a neutral solution? [1 marks]
 - (a ii)** Name one acid found in the human stomach. [1 marks]
 - (b)** Explain why sodium chloride conducts electricity in its molten state but not in its solid state. [3 marks]
- 12.** Draw a labelled diagram of the human digestive system. [5 marks]